

R Niranjan

Coimbatore, Tamil Nadu, India • +91 9345988554 • Niranjan.r.dev@proton.me

Portfolio: niranjanr.in • LinkedIn: linkedin.com/in/niranjan-r-879968365

GitHub: github.com/TGvenomYT

PROFESSIONAL SUMMARY

Software developer building at the intersection of generative AI, real-time voice infrastructure, DevOps automation, and applied security. Ships LLM-powered systems end to end — voice agents on LiveKit, autonomous operations and reconnaissance agents, and a unified AI tool gateway built on the Model Context Protocol — on a foundation of Python and self-hosted Linux infrastructure. Active open-source contributor and organizer in the Coimbatore tech community.

EDUCATION

B.Tech, Artificial Intelligence & Data Science — KPR Institute of Engineering and Technology (KPRIET)

Coimbatore, Tamil Nadu • Expected 2030 • First-year student

TECHNICAL SKILLS

- **Programming Languages:** Python, C, Shell/Bash, HTML, Markdown
- **AI & Machine Learning:** Google Gemini, Ollama, LangChain, LangGraph, LangSmith, Model Context Protocol (MCP)
- **Voice & Audio:** LiveKit, Cartesia Sonic (TTS), ElevenLabs, Kokoro TTS
- **DevOps & Infrastructure:** Docker & Docker Compose, PM2, Grafana, Loki, Cloudflare Tunnels, Tailscale, GitHub Copilot
- **Backend & Data:** Supabase / PostgreSQL, REST APIs, self-hosted services, GitHub Actions (self-hosted runner)
- **Operating Systems:** Windows, macOS, Linux (all major distributions)
- **Tools & Platforms:** AppFlowy, Lovable, Figma, Canva

WORK EXPERIENCE

Software Automation & ML Development Intern — HummingBird Digital Pvt. Ltd.

Apr 2026 – Present (~5 months)

- Architected and deployed automated voice-calling and booking-reconfirmation workflows for the HotelCallingAgent-Reconfirmation product.
- Cut per-call vendor cost by moving from Plivo to an in-house ElevenLabs voice agent, while retaining Exotel telephony across 15 provisioned numbers.
- Built real-time conversational voice agents on LiveKit's open-source infrastructure, integrating Cartesia Sonic for low-latency text-to-speech.
- Developed an autonomous DevOps agent that manages Docker containers and PM2 processes from natural-language commands.
- Contributed to an automated hotel-mapping solution.
- Built Rotary Circle, a B2B lead-generation marketplace for Rotary RID 3206.

SELECTED PROJECTS

- **AgentKit — Unified AI Tool Gateway:** Designed a 14-tool gateway that exposes DevOps, terminal, and personal-assistant capabilities to any AI agent through three interfaces — a CLI, Ollama function-schemas, and a Model Context Protocol (MCP) server; verified against local Gemma models.

- **Jarvis — Offline Voice Assistant:** Built a fully offline voice assistant on LiveKit with a Flutter client and Python backend, running local Gemma models via Ollama/MLX; tuned model inference to eliminate thinking-token latency for real-time conversation.
- **SafeSphere (Smart India Hackathon 2026):** Post-disaster victim-location and rescue-coordination platform using BLE mesh networking; built the working proof-of-concept as part of the team, including a companion Local Link module and live Central Water Commission river-gauge integration so flood alerts arm from real-time hydrology data.
- **Self-Hosted Homelab & Infrastructure:** Operate a Linux server running Docker, Grafana/Loki observability, Cloudflare Tunnels, and a Tailscale mesh VPN; host services including a self-hosted Supabase stack, a CI runner, and a React + Express monitoring dashboard.
- **Cortex — Personal AI Daemon:** A local-LLM personal assistant with scheduled spoken briefings, backed by a self-hosted Supabase database.
- **Campus Mesh Chat:** Offline peer-to-peer messaging for the KPRIET campus built on a BitChat (BLE mesh) fork — Android-first — enabling communication with no internet or cellular coverage.
- **Cross-Platform App Release Engineering:** Shipped Android apps end to end — build, code-signing, permissions, and release — and set up an iOS build-and-signing pipeline for Flutter applications.
- **KVM CMS:** Content management system for KVMTC, Brindavan Nagar, Cheran Ma Nagar, Coimbatore.
- **AI-Assisted Web Development:** Rapidly prototyped and shipped web projects with the Lovable AI application builder, including an ad-free music-streaming aggregator; deployed a personal portfolio (niranjanr.in) on GitHub Pages with a custom domain.

SECURITY ENGINEERING

- **Blackwall — Automated Recon Agent:** Built a homelab-scoped reconnaissance and enumeration agent (nmap, ffuf, web probing) orchestrated by an LLM over SSH for authorized security testing.
- **Guardian — MCP Endpoint Defense:** Implemented request logging, rate-limiting, and automated banning of crawlers and probes to protect a public MCP endpoint, exposed as agent-callable tools.
- **Homelab Security Audit:** Assessed a self-hosted server's attack surface — surfacing firewall gaps around Docker-published ports and over-privileged containers — and hardened the configuration.
- **Personal Breach Monitor:** Credential-exposure checker that audits one's own email and passwords against public breach corpuses (XposedOrNot, HIBP).

SYSTEMS & NETWORKING

- **Linux Administration:** Manage containerized services, process supervision (PM2/systemd), and CUPS printing across the homelab, including AppArmor and driver troubleshooting.
- **Network Configuration:** Configured LAN DNS and static addressing on a home router, plus Tailscale mesh and Cloudflare Tunnel networking to reach self-hosted services securely.

COMMUNITY & LEADERSHIP

- **Microsoft Learn Student Ambassador** (Aug 2026 – Present)
- **Secretary, Rotary Manchester Annette's Club** (RID 3206), 2026–2027
- **Member, Global AI Community** — Coimbatore Chapter
- **BrewingSec CyberDev Summit (Jul 2026)** — hands-on cybersecurity workshop by BlackPerl DFIR, KGiSL Institute of Technology, Coimbatore
- **Pie & AI: Coimbatore (Jul 2026)** — “AI Agents for Everyone” session on practical autonomous agents